# Continuity of the sectorial bounded holomorphic calculus

A full answer to the open problem in arXiv:1101.0067

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**Status.** High-confidence candidate full proof, pending human review. The only normalization needed is made explicit below: since the source gives $f$ only on the selected sector, $f(A)$ means $f$ on that spectral sector and zero on the other one. For $f \equiv 1$ this is exactly the source's sectorial projection.

## The source question

Let $M$ be closed, $E \to M$ Hermitian, $m > 0$, and let $\Gamma_+$ be the fixed two-ray contour of the source. It divides the plane into sectors $\Lambda_+$ and $\Lambda_-$. The source proves continuity of $A \mapsto P_{\Gamma_+}(A)$ on $\mathrm{Ell}^m_{\Gamma_+}(M, E)$ in its topology $\mathcal{T}$, and asks whether the theorem extends to bounded holomorphic functions.

> **Remark 5.1.** We note in passing that in the first three lines of the proof of Proposition 3.1 it is decisively used that the integrand is *homogeneous* in $\lambda$. This is an obstruction against an immediate generalization of our perturbation result to general holomomorphic functions of $A$ in an $H_\infty$–calculus style (*cf., e.g.,* [7]). To explain this a bit more, let $f$ be a function, which is holomorphic and bounded in a neighborhood of the closure of the sector encircled by $\Gamma_+$. This is not quite the situation of the $H_\infty$ calculus, since there are also eigenvalues on the left of the contour. Nevertheless, one might hope that similarly to loc. cit. one can prove that $f(A)$ is a bounded operator. Still there is no immediate analogue of Proposition 3.1 in this case and we leave it as an intriguing open problem whether the Perturbation Theorem 1.2 carries over to $f(A)$ instead of $P_{\Gamma_+}(A)$.

The exact page is bundled as `source_excerpt_open_problem_page_17.pdf`.

Fix disjoint open neighborhoods $U_\pm$ of the closed sectors and $f \in H^\infty(U_+)$. Put

$$F(z) = \begin{cases} f(z), & z \in U_+, \\ 0, & z \in U_-. \end{cases}$$

Thus $F$ is holomorphic on the disconnected neighborhood $U_+ \cup U_-$. Write $f_{\Gamma_+}(A) := F(A)$ for the corresponding sectorial holomorphic calculus, initially for functions decaying at zero and infinity and then by the standard bounded-calculus convergence procedure.

**Theorem 1.** *For every $s \in \mathbb{R}$ and every fixed $f \in H^\infty(U_+)$, the operator $f_{\Gamma_+}(A)$ is bounded on $H^s(M; E)$ and*

$$\mathrm{Ell}^m_{\Gamma_+}(M, E) \longrightarrow \mathcal{B}(H^s(M; E)), \qquad A \longmapsto f_{\Gamma_+}(A),$$

*is continuous in the source topology $\mathcal{T}$. Locally in $A$ one has $\|f_{\Gamma_+}(A)\|_{s,s} \le C \|f\|_\infty$.*

## The two ingredients

Write the topology exactly as the source does:

$$A = \mathrm{Op}(a_m) + B, \qquad a_m \in C^\infty(S^*M; \mathrm{End}(\pi^*E)), \qquad B \in \Psi^{m-1}(M; E).$$

Then $A_j \to A$ in $\mathcal{T}$ means $a_m(A_j) \to a_m(A)$ in $C^\infty$ and

$$\|B(A_j) - B(A)\|_{k+m-1,k} \longrightarrow 0 \quad \text{for every integer } k. \tag{1}$$

Interpolation gives the analogous statement at all real Sobolev indices.

The first ingredient is the symbol-level bounded holomorphic calculus of Bilyj–Schrohe–Seiler. Their Theorems 3.2 and 3.5 prove that bounded holomorphic functions of a sectorially parameter-elliptic symbol are symbols of order zero, with every symbol seminorm and the induced operator norm bounded by $C\|f\|_\infty$. Their proof uses local Cauchy contours around the pointwise symbol spectrum. Consequently it applies componentwise to $F$: the contours around the $+$ component contribute $f$, while those around the $-$ component contribute zero.

**Lemma 1** (frozen lower-order family)**.** *Fix $A_0 = \mathrm{Op}(a_0) + B_0$. For $a$ in a sufficiently small $C^\infty$ neighborhood of $a_0$, set $C(a) = \mathrm{Op}(a) + B_0$. Then*

$$a \longmapsto f_{\Gamma_+}(C(a)) \in \mathcal{B}(H^s)$$

*is continuous and locally bounded by $C\|f\|_\infty$.*

*Justification.* Uniform separation of the principal spectrum from the two cutting rays persists in a $C^0$ neighborhood. On the unbounded part, the cited parameter-dependent construction gives an order-zero symbol; the resolvent identity on its local Cauchy contours makes each symbol seminorm continuous in $a$. The finitely many bounded spectral points are enclosed by a compact Dunford contour, where ordinary resolvent continuity applies. Quantization $\Psi^0 \to \mathcal{B}(H^s)$ is continuous. This proves the lemma. Notice that the lower-order symbol is fixed, so no topology stronger than $\mathcal{T}$ is being silently imposed.

The second ingredient is the extra resolvent produced by comparing two operators. Put

$$\delta := \min\{1, 1/m\}.$$

**Lemma 2** (integrable resolvent difference)**.** *Let $A, C$ lie in a sufficiently small $\mathcal{T}$-neighborhood, have the same principal symbol, and put $D = A - C \in \Psi^{m-1}$. On either unbounded ray of $\Gamma_+$,*

$$\|(A-\lambda)^{-1} D (C-\lambda)^{-1}\|_{s,s} \le C_s\, q_s(D) |\lambda|^{-1-\delta}, \tag{2}$$

*where*

$$q_s(D) = \begin{cases} \|D\|_{s+m-1,s}, & m \ge 1, \\ \|D\|_{s,s+1-m}, & 0 < m < 1. \end{cases}$$

*Both are continuous seminorms of $\mathcal{T}$.*

*Proof.* The source's minimal-growth estimate is

$$\|(T-\lambda)^{-1}\|_{u,u+p} \le C|\lambda|^{-1+p/m}, \qquad 0 \le p \le m, \tag{3}$$

locally uniformly in $T$. If $m \ge 1$, apply (3) with $p = m-1$ to the right resolvent, then $D : H^{s+m-1} \to H^s$, and apply (3) with $p = 0$ to the left resolvent. This gives $|\lambda|^{-1/m}|\lambda|^{-1}$. If $0 < m < 1$, use both resolvents on their fixed Sobolev spaces with decay $|\lambda|^{-1}$ and use the smoothing map $D : H^s \to H^{s+1-m}$ between them. This gives $|\lambda|^{-2}$. These are exactly (2). ∎

## Proof of Theorem 1

Fix $A_0 = \mathrm{Op}(a_0) + B_0$. For nearby $A = \mathrm{Op}(a) + B$, freeze its lower-order part at the base point:

$$C = C(a) := \mathrm{Op}(a) + B_0, \qquad D := A - C = B - B_0. \tag{4}$$

Then $C \to A_0$ in the ordinary full-symbol topology, while (1) says $q_s(D) \to 0$.

First take a holomorphic $f$ with the usual decay at zero and infinity. The resolvent identity gives, with the standard orientation,

$$f_{\Gamma_+}(A) - f_{\Gamma_+}(C) = \frac{1}{2\pi i} \int_{\Gamma_+} f(\lambda)(A-\lambda)^{-1} D (C-\lambda)^{-1}\, d\lambda. \tag{5}$$

By Lemma 2, the two rays contribute at most

$$C_s \|f\|_\infty q_s(D) \int_R^\infty r^{-1-\delta}\, dr.$$

The connecting compact arc contributes the same type of bound by uniform resolvent continuity. Hence

$$\|f_{\Gamma_+}(A) - f_{\Gamma_+}(C)\|_{s,s} \le K_s \|f\|_\infty q_s(D). \tag{6}$$

Choose the standard bounded holomorphic regularizers $\rho_n$, decaying at zero and infinity, with uniformly bounded sup norms and $\rho_n \to 1$ locally uniformly on both sectors. Estimate (6), uniform in $n$, and the bounded calculus convergence lemma pass (5)–(6) from $\rho_n F$ to every bounded $F$. This proves boundedness of $f_{\Gamma_+}(A)$ and retains (6).

Finally,

$$\begin{aligned} \|f_{\Gamma_+}(A) - f_{\Gamma_+}(A_0)\|_{s,s} \le K_s \|f\|_\infty q_s(B - B_0) \\ + \|f_{\Gamma_+}(C(a)) - f_{\Gamma_+}(C(a_0))\|_{s,s}. \end{aligned}$$

The first term tends to zero by (1), and the second by Lemma 1. This proves continuity at the arbitrary base point $A_0$. The same estimates give the local norm bound. $\square$

## Proof intuition and scope

The source's direct integral has only one resolvent and therefore decays like $|\lambda|^{-1}$, exactly at the divergent endpoint. Freezing the lower order part separates the two pieces of $\mathcal{T}$. Principal-symbol variation is handled by the established bounded symbol calculus; lower-order variation is handled by a resolvent difference, which automatically contains two resolvents. The order gap of one supplies the additional $|\lambda|^{-\delta}$ that makes the contour integral converge.

The result answers Remark 5.1 under the canonical sectorial interpretation. It does not assert continuity when the function $f$ itself varies without a uniform holomorphic domain, nor does it weaken the source's fixed-cut and minimal-growth hypotheses. Exact-ID, title, citation, and official-arXiv searches through 12 August 2026 found no later paper explicitly closing this perturbation question; this is a bounded novelty audit, not a priority claim.

## References


**[1]** B. Booß–Bavnbek, G. Chen, M. Lesch, and C. Zhu, *Perturbation of Sectorial Projections of Elliptic Pseudo-differential Operators*, arXiv:1101.0067, Theorem 1.2, Definition 2.7, Lemma 4.7, and Remark 5.1.

**[2]** O. Bilyj, E. Schrohe, and J. Seiler, *$H_\infty$-calculus for hypoelliptic pseudodifferential operators*, arXiv:0901.3160, Theorems 3.2 and 3.5.